What is C language?
Answer: C is a general-purpose, procedural computer programming language. It is known for its efficiency and low-level memory manipulation, and it is widely used for developing operating systems, compilers, and embedded systems.
2. What are the applications of C language?
Answer: C language has a wide range of applications, including:
Operating Systems: C is used to develop operating systems like Unix, Linux, and Windows.
Embedded Systems: It is a preferred language for programming microcontrollers and embedded systems due to its efficiency.
Compilers and Interpreters: Many compilers and interpreters for other programming languages are written in C.
Database Management Systems: C is used in the development of database systems like MySQL.
System Utilities: It is used to create system utilities and tools.
3. What is a Variable?
Answer: A variable is a named storage location in a computer's memory that holds data. It is used to store and manipulate data in a program. Each variable has a specific data type, which determines the type of values it can hold.
4. What are different datatypes of a C program?
Answer: C language supports several fundamental data types, which can be broadly categorized as follows:
int: Used to store integer values (e.g., 10, -5).
float: Used to store single-precision floating-point numbers (e.g., 3.14).
double: Used to store double-precision floating-point numbers (e.g., 3.14159265).
char: Used to store a single character (e.g., 'A', 'b').
void: An incomplete type that represents the absence of a value.
5. What is a Format specifier?
Answer: A format specifier is a placeholder used in formatted input and output functions (like printf() and scanf()) that tells the compiler what type of data to expect or display. It begins with a percent sign (%).
Common format specifiers include:
%d: Used for int data type.
%f: Used for float data type.
%c: Used for char data type.
%s: Used for strings.
%lf: Used for double data type.